



DNS Live Lab Worksheet

“Control the Internet Using DNS”

Duration: 30 minutes

Objective: Understand how DNS works by observing, testing, breaking, and fixing it on your own system.



Instructions

- Complete all parts in order
- Use only free tools (Command Prompt / Terminal)
- Take and attach screenshots where required
- Answer all validation questions

PART 1 — DNS Lookup (The Phonebook)

Task:

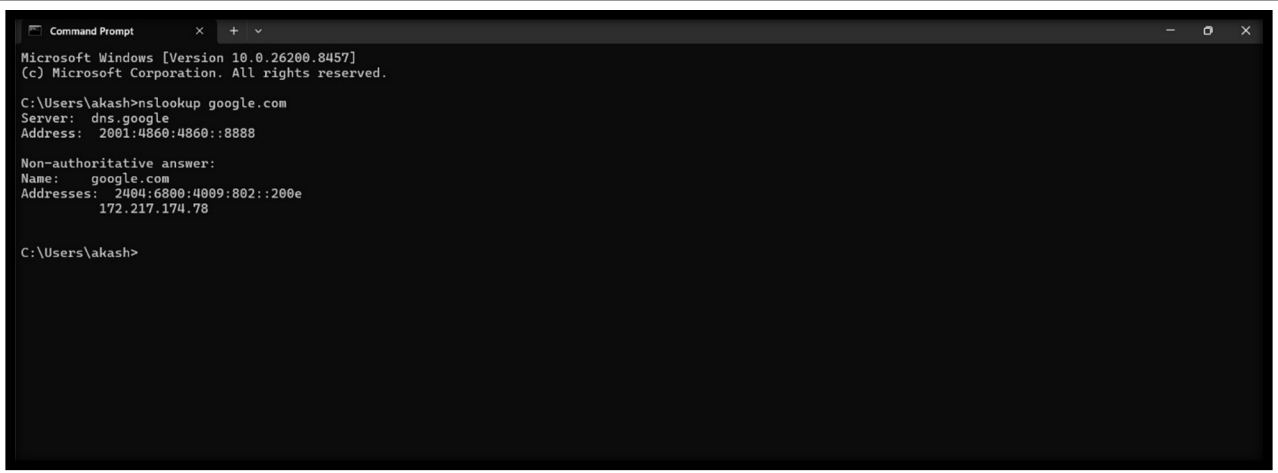
Run the following command:

```
nslookup google.com
```

Record Your Observations:

1. IP address of google.com: 172.217.174.78
2. DNS server used: (dns.google)2001:4860:4860::8888

 Screenshot Required: Yes (attach your screenshot here)



```
Command Prompt
Microsoft Windows [Version 10.0.26200.8457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\akash>nslookup google.com
Server: dns.google
Address: 2001:4860:4860::8888

Non-authoritative answer:
Name: google.com
Addresses: 2404:6800:4009:802::200e
          172.217.174.78

C:\Users\akash>
```

PART 2 — Compare Different DNS Servers

Task:

Run:

```
nslookup google.com 8.8.8.8
nslookup google.com 1.1.1.1
```

Answer:

3. Do both DNS servers return the same IP?

No

4. What does this indicate about DNS systems?

Different DNS servers can return different IP addresses for the same website because large websites like Google use many servers worldwide. DNS servers may provide the IP address of the server that is closest or most suitable for the user improving performance and reliability.

 Screenshot Required: Yes (attach your screenshot here)

```
C:\Users\akash>nslookup google.com 8.8.8.8
Server:  dns.google
Address:  8.8.8.8

Non-authoritative answer:
Name:     google.com
Addresses: 2404:6800:4009:80e::200e
          142.251.222.78

C:\Users\akash>nslookup google.com 1.1.1.1
Server:  one.one.one.one
Address:  1.1.1.1

Non-authoritative answer:
Name:     google.com
Addresses: 2404:6800:4007:830::200e
          142.251.43.238

C:\Users\akash>
```



PART 3 — Explore Multiple Domains



Task:

Run the following commands:

```
nslookup facebook.com
nslookup youtube.com
nslookup amazon.com
```



Answer:

5. Write one IP address for each domain:

- facebook.com: 163.70.140.35
- youtube.com: 142.250.76.206
- amazon.com: 98.87.170.71

6. Do all websites have the same IP?

No

7. What does this tell you about how the internet is structured? (Describe in your own words)

Different websites are hosted on different servers and networks, so they have different IP addresses. DNS translates domain names into these IP addresses, allowing users to access the correct website. This shows that the Internet is made up of many interconnected servers rather than one central system.



Screenshot Required: Yes (attach your screenshot here)

```
C:\Users\akash>nslookup facebook.com
Server: Unknown
Address: 192.168.142.206

Non-authoritative answer:
Name: facebook.com
Addresses: 2a83:2880:f185:85:face:b00c:0:25de
163.70.140.35

C:\Users\akash>nslookup youtube.com
Server: Unknown
Address: 192.168.142.206

Non-authoritative answer:
Name: youtube.com
Addresses: 2404:6800:4000:101d:8b
2404:6800:4000:101d:8b
2404:6800:4000:101d:be
2404:6800:4000:101d:5d
142.250.76.206

C:\Users\akash>nslookup amazon.com
Server: Unknown
Address: 192.168.142.206

Non-authoritative answer:
Name: amazon.com
Addresses: 98.87.170.71
98.87.170.74
98.82.161.185
```



PART 4 — DNS Server Comparison



Task:

Run:

```
nslookup youtube.com 8.8.8.8  
nslookup youtube.com 1.1.1.1
```



Answer:

8. Do different DNS servers return the same IP?

No

9. What does this indicate about DNS globally?

Different DNS servers can return different IP addresses for the same domain, showing that DNS is distributed worldwide and uses load balancing and geographic optimization to provide reliable and efficient access to websites.



Screenshot Required: Yes (attach your screenshot here)

```
C:\Users\akash>nslookup youtube.com 8.8.8.8  
Server: dns.google  
Address: 8.8.8.8  
  
Non-authoritative answer:  
Name: youtube.com  
Addresses: 2404:6800:4009:809::200e  
142.250.207.174  
  
C:\Users\akash>nslookup youtube.com 1.1.1.1  
Server: one.one.one.one  
Address: 1.1.1.1  
  
Non-authoritative answer:  
Name: youtube.com  
Addresses: 2404:6800:4000:100c::5d  
2404:6800:4000:100c::be  
2404:6800:4000:100c::5b  
2404:6800:4000:100c::88  
142.250.143.91  
142.250.143.136  
142.250.143.190  
142.250.143.93
```